

Multimodal Image-Based Phenotyping for Early Selection of Potential Heat-Tolerant Variants in Irradiated *Cnidium officinale*

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BACKGROUND

Climate change induced high temperatures and extreme weather event s are increasingly threatening the stable cultivation and production of *Cnidium officinale*, leading to a gradual decline in the number of farms and the overall production base. In addition, because *C. officinale* is a vegetatively propagated crop with limitations in conventional crossbreeding, radiation mutation breeding is needed to expand its genetic diversity and develop novel germplasm with enhanced climate adaptability.

BREEDING BOTTLENECK

Vegetative propagation and the absence of seed production severely restrict conventional crossbreeding. Radiation mutation breeding offers a practical approach to broadening genetic diversity.

OBJECTIVE

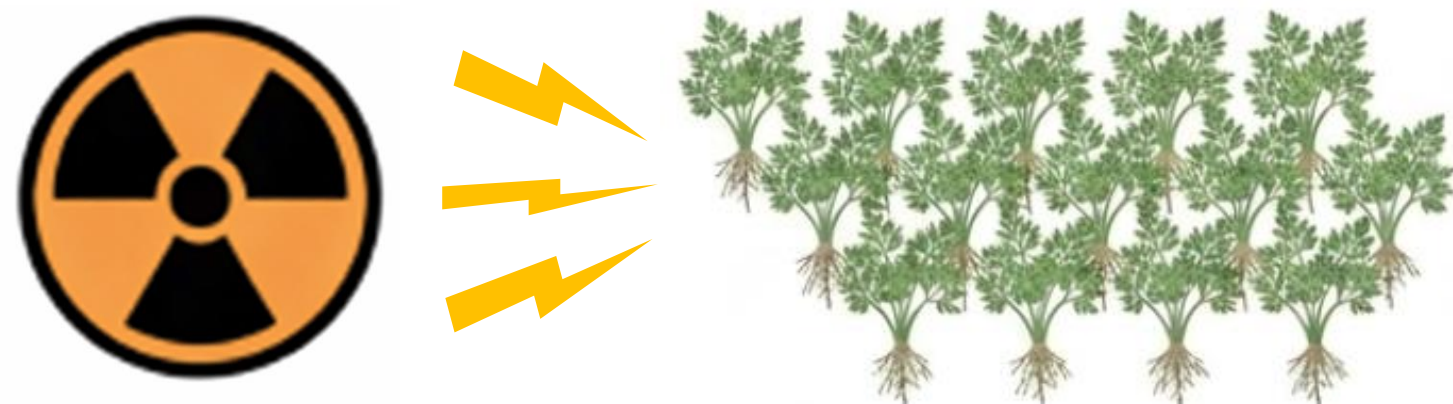
To establish a stepwise screening framework that integrates color, thermal, vegetation and growth signals to prioritize climate-adaptive variants from a large irradiated population.

MATERIALS & MEASUREMENTS

01

Radiation population

Gamma-ray and electron-beam treated rhizomes with non-irradiated controls



02

Color phenotype

Purple pigmentation, anthocyanin-related indices and visible variants



03

Thermal phenotype

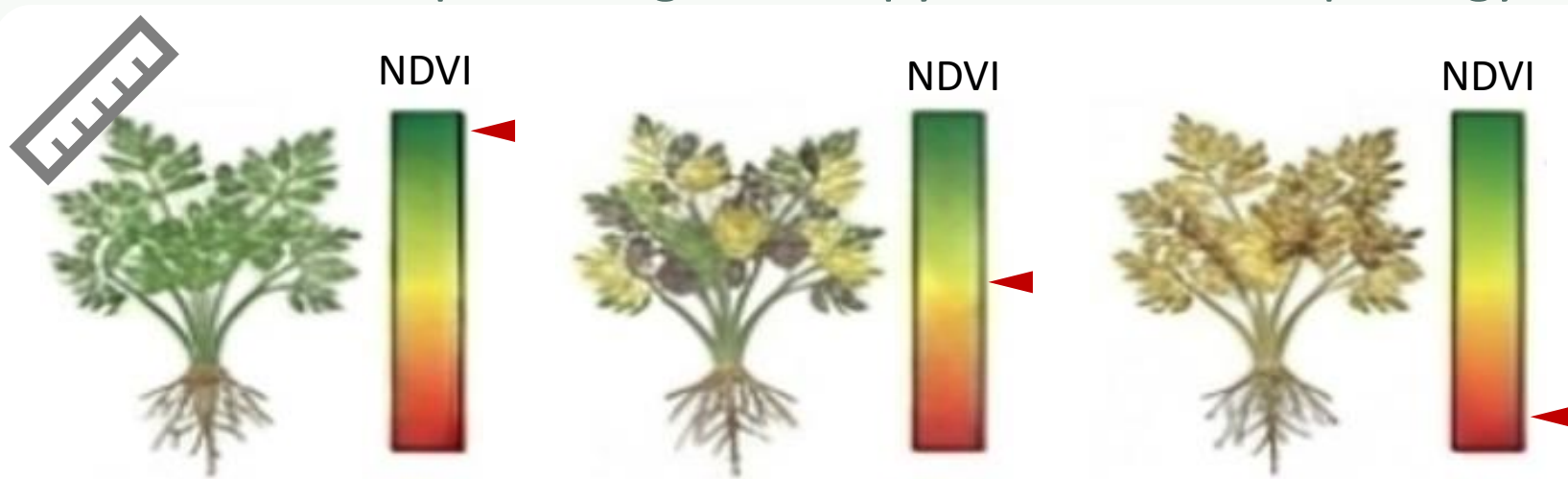
Time-course canopy temperature from morning to nighttime



04

Plant performance

NDVI, survival, plant height, canopy width and morphology



STEPWISE SELECTION FRAMEWORK

Each modality evaluates a different aspect of the plant phenotype, and candidates are prioritized by integrating complementary signals.

COLOR SIGNAL

Mutation potential

Early purple pigmentation identifies plants with an increased probability of radiation-induced phenotypic variation.

THERMAL SIGNAL

Heat response

Diurnal canopy-temperature dynamics reveal cooling capacity, heat accumulation and post-stress recovery.

VEGETATION SIGNAL

Plant vigor

NDVI helps distinguish favorable cooling from low temperature caused by weak or sparse growth.

GROWTH SIGNAL

Agronomic relevance

Survival and growth traits remove severely damaged plants and retain viable, useful candidates.

ENVIRONMENTAL NORMALIZATION

$$\Delta T = (\text{Canopy temp}) - (\text{Air temp})$$

Interpreted with soil moisture, humidity, canopy architecture and measurement time.

INTEGRATED OUTPUT

High-priority candidate variants

Plants combining mutation potential, favorable thermal behavior, maintained NDVI and acceptable growth advance to validation.

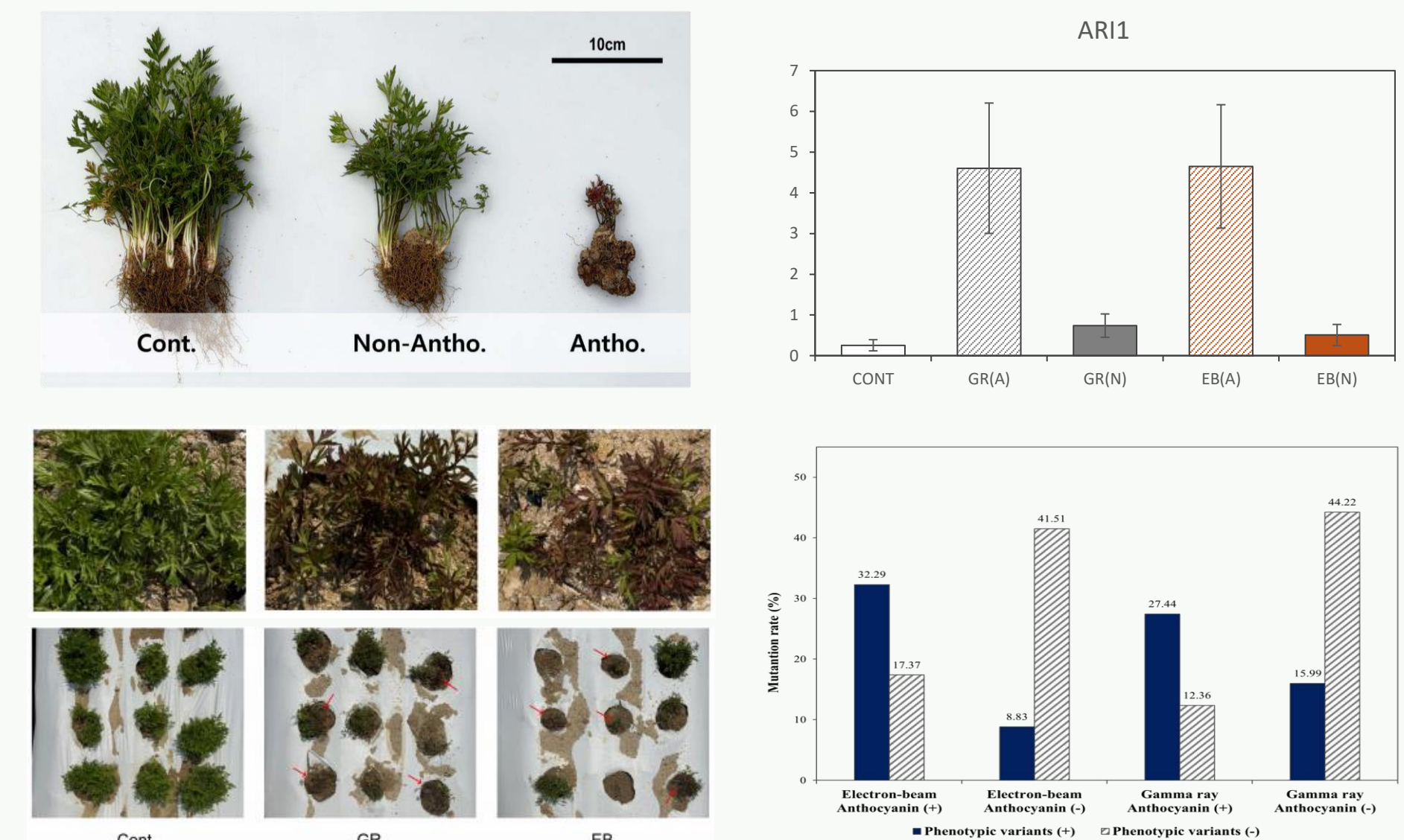
INTERPRETATION

Anthocyanin is a selection indicator—not proof of a useful mutation.

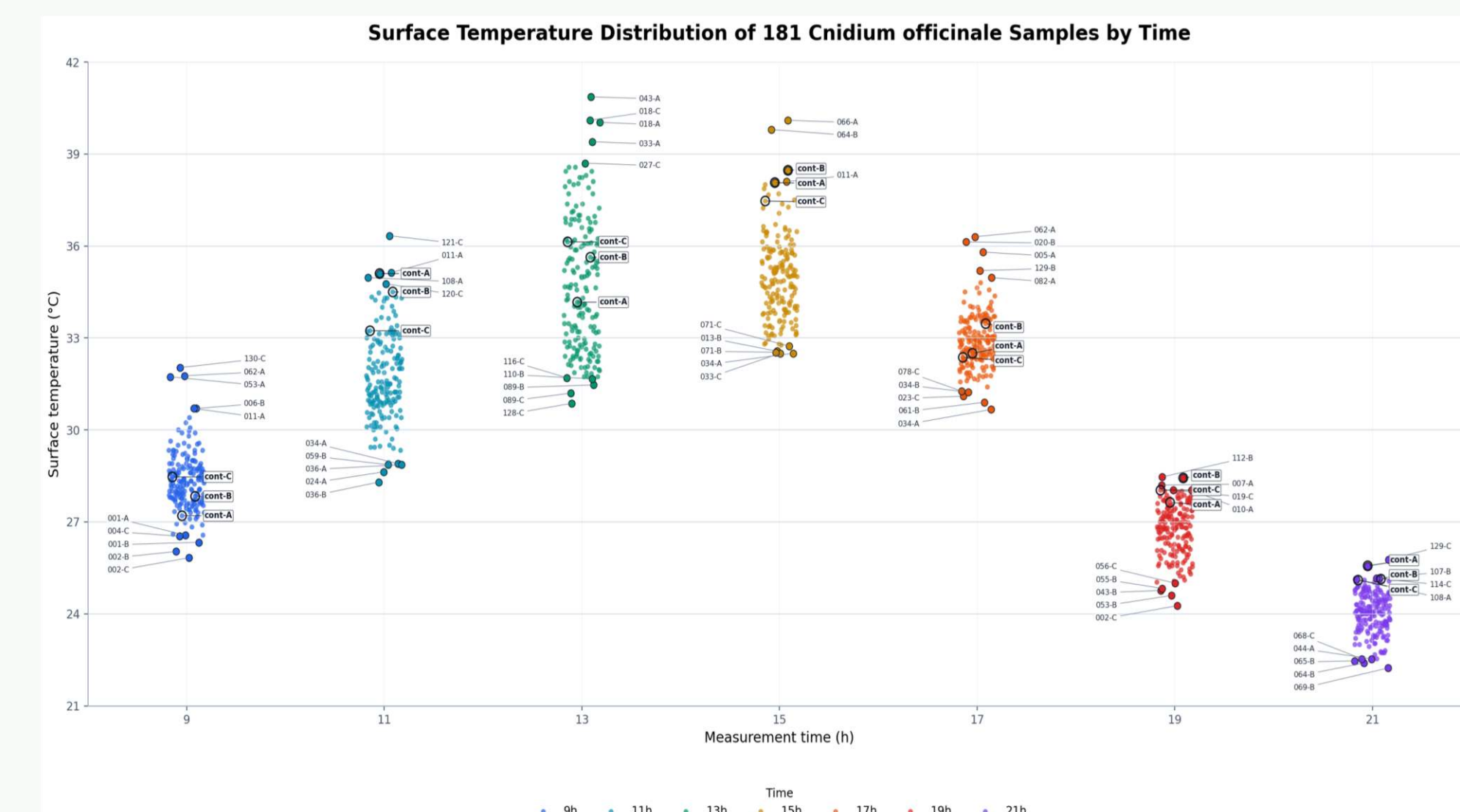
Pigmentation supports early enrichment of the population, while heritability and agronomic value require subsequent evaluation.

KEY FINDINGS

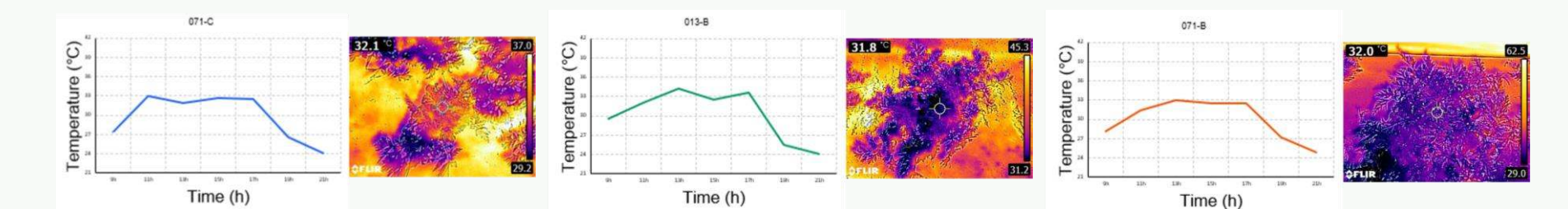
01 (Color Signal) Mutation potential



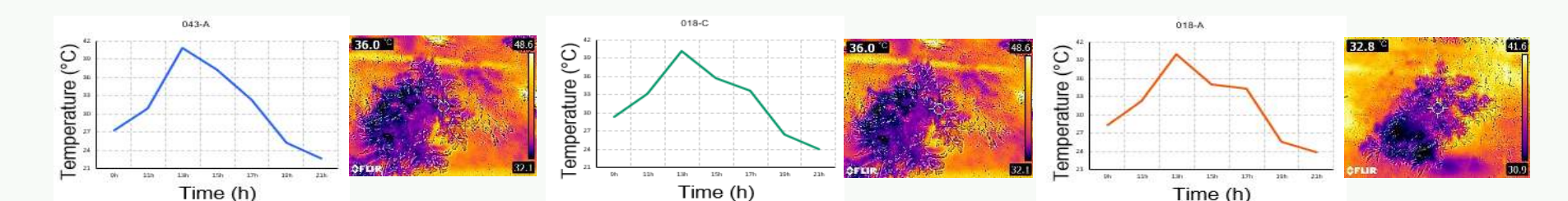
02 (Thermal Signal) Heat response



Cool canopy temp



Hot canopy temp



OUTLOOK

The framework can be extended beyond heat stress to drought, waterlogging and disease resistance. Automated segmentation and predictive modeling could enable rapid, scalable selection in medicinal crops.

KEY MESSAGE

Multimodal signals reduce a large irradiated population to a smaller, biologically credible set of climate-adaptive candidates.